INCH-POUND

MIL-A-23595D(AS)
14 APRIL 1995
SUPERSEDING
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15 June 1988

PERFORMANCE SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for Audio Frequency Amplifiers used in Aircraft Intracommunication and Intercommunications Systems. The amplifiers are in-line type mounted on the helmet or in the oxygen mask. The amplifier shall be designed to operate with an input from a 5 ohm dynamic microphone and provide output to an intercommunication system such as the AN/AIC-14A.

1.2 Part numbers. Part numbers for amplifiers furnished under this specification shall be as follows: M23595/X-Y with "X" being the specification sheet number and "Y" being any variation of the amplifier on a specific specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander Systems Standardization Department Code 4.1.11 Naval Air Warfare Center, Aircraft Division, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-P-416	Plating Cadmium (Electrodeposited)

MILITARY

MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-E-6051	Electromagnetic Compatibility Requirements Systems
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts): Packaging of
MIL-T-18303	Test Procedures; Production, Acceptance, and Life for Aircraft Electronic Equipment, Format for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Requirements For The Control of Electromagnetic Interference Emissions and Susceptibility
MIL-STD-462	Measurement of Electromagnetic Interference Characteristics
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1285	Marking of Electrical and Electronic Parts

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MIL-HDBK-235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment Subsystems and Systems
MIL-HDBK-237	Electromagnetic Compatibility Management Guide for Platforms, Systems and Equipment

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawing, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAIR AD 1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment

(Copies of Naval Air Systems Command documents may be obtained from Commander, Naval Air Systems Command, Code 4.1C, 1421 Jefferson Davis Highway, Arlington, VA 22202-5511.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the detail specification, the latter shall govern. (If a specific requirement specified herein is not required for an item, it shall be so indicated on the specification sheet (e.g., Shock - N/A).)

3.2 Qualification. Amplifiers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.4 and 6.3). In the event that no vendors are listed on the proper QPL, then the procuring activity shall require First Article testing equivalent to Qualification testing prior to the delivery of any production units. The qualifying activity shall have cognizance over any First Article performed under this specification.

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3.2.1 First article inspection. Amplifiers furnished under this specification not covered by specification sheets or not qualified at the time set for the opening of bids, shall be subjected to the tests specified in 4.4 (see 4.4.5).

3.3 Materials.

3.3.1 Selection of parts and materials. Selection of parts and materials shall be in accordance with MIL-E-5400.

3.3.1.1 Approval of nonstandard parts and materials. Approval of nonstandard parts shall be in accordance with MIL-E-5400, Category III.

3.3.2 Fungus. The amplifiers shall conform to Requirement 4, fungus inert materials, of MIL-STD-454.

3.3.3 Toxicity. Materials used in the design of the amplifier shall be non-toxic in accordance with MIL-STD-454, Requirement 1.

3.3.4 Flammable. Materials which will support combustion shall not be used. Definitions and restrictions on the use of flammable materials shall be in accordance with MIL-STD-454, Requirement 3.

3.3.5 Metals. Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

* 3.3.6 Deleted

* 3.3.7 Deleted

3.3.8 Aluminum alloy parts. Aluminum alloy parts shall be anodically treated in accordance with MIL-C-5541. To provide electrical continuity, the aluminum oxide film formed by this treatment shall be removed from the actual contact area of all surfaces that act as path for electrical current and from the areas under screws and nuts used for assembly or mounting purposes.

3.3.9 Steel parts. Steel parts shall be cadmium plated in accordance with QQ-P-416 Type II or III.

3.3.10 Insulating materials. Electrical insulating materials shall be in accordance with MIL-STD-454, Requirement 11.

3.3.11 Embedment (potting). Embedment compounds shall be in accordance with MIL-STD-454, Requirement 47.

* 3.3.12 Soldering. Soldering shall be in accordance with MIL-STD-2000.

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* 3.4 Design and construction. The amplifier shall conform with all the applicable requirements of MIL-E-5400, Class I, for design and construction, unless otherwise specified herein or in the applicable specification sheet. The interior of the amplifier housing shall be potted to prevent corrosion of internal components due to environmental exposure.

3.4.1 Form factor. The form factor shall be as specified in the applicable detail specification sheet.

3.4.1.1 Weight. The weight of the amplifier shall not exceed 0.75 ounce.

3.4.1.2 Color. The color of the amplifier shall be black.

3.4.2 Connections. The connections shall be as specified in the applicable detail specification sheet.

3.4.3 Housing. The housing shall be constructed so that it shall be capable of demonstrating conformance to all the requirements and tests of this specification. The preferred construction shall be of aluminum alloy. However, a molded plastic case may be used if suitable shielding is provided.

3.4.3.1 Pin stress. The amplifier output pins shall be capable of withstanding a force of 45 pounds applied to the pins in a plane 90 degrees to their projection when tested in accordance with 4.7.23 without cracking, splitting, separating or causing any other effect that would render the amplifier unserviceable.

3.4.4 DC power requirements. The amplifier shall be designed to accept the DC supply voltage at the output terminals. The amplifier shall provide specified performance with any supply voltage in the range from 22.0 to 30.0 volts DC. Under low voltage conditions, 12.0 to 22.0 volts DC, the gain shall be as specified in 3.5.2.3. The amplifier shall not be polarity sensitive.

3.5 Electrical characteristics.

3.5.1 Function. The amplifier shall provide amplification of signals derived from the output of a 5 ± 1 ohms dynamic-type microphone. Amplified microphone signals shall be capable of delivering rated output voltage into 100 ohms ± 1% at 1000 Hz. The amplifier shall be powered as described in 3.4.4.

3.5.2 Gain.

3.5.2.1 Gain under standard conditions. With a nominal input voltage of 250 microvolts RMS, the amplifier shall have a voltage gain of 61 ± 1 dB at 1000 Hz when tested in accordance with 4.7.1 and 4.3.1. The response between 50 and 10,000 Hz shall be within the limits of Figure 1. The gain at 100,000 Hz shall be no more than 20 dB. The voltage gain shall not vary more than 0.5 dB from the gain established at 27.5 volts when the DC supply voltage is varied between 22 and 30 volts.

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3.5.2.2 Gain under temperature-altitude conditions. When the amplifier is subjected to the conditions of 4.7.22.1 and is operated between the extremes of -54 to -35 degrees C, the gain may decrease by a maximum of 6 dB referenced to the output under standard conditions at 1000 Hz. When operated between -34 and +55 degrees C, the gain may not vary by more than ±3 dB when referenced to the output under standard conditions at 1000 Hz.

3.5.2.3 Gain under low DC voltage conditions. The amplifier's 1000 Hz gain shall be not more than -12 dB, referenced to the output obtained with a power source of 27.5 volts, when the power source voltage is varied between the limits of 12 and 22 volts.

3.5.2.4 Gain stabilization time. The amplifier gain of 61 ± 1 dB at 1000 Hz shall be attained within 1 dB in 100 milliseconds or less after the application of DC input power. Under low voltage conditions, 12 to 22 volts, the amplifier shall attain the gain required by 3.5.2.3 within 150 milliseconds or less after the application of DC input power.

3.5.3 Impedance. The input impedance of the amplifier shall be 5 + 10, -2 ohms at 1000 Hz. The amplifier shall provide rated output voltage into a 100 ohm (±1%) resistive load.

3.5.4 Voltages.

* 3.5.4.1 Signal input voltage. The nominal signal input voltage shall be 250 microvolts RMS with a tolerance of ±5%. This input voltage shall be used for test purposes unless otherwise specified.

3.5.4.2 Bias voltage. The nominal bias voltage shall be 27.5 volts DC with a tolerance of ±5%. The nominal operating range shall be 22 to 30 volts DC. The bias voltage refers to the voltage of the DC power supply of Figure 2 unless otherwise specified. The nominal operating range shall apply for test purposes unless otherwise specified.

3.5.5 Linearity. When a 1000 Hz input signal is varied between 50 and 500 microvolts, the output shall maintain a constant gain of 61 ± 1 dB over the nominal bias voltage range of 22 to 30 volts DC.

* 3.5.6 Deleted.

3.5.7 Distortion. The total harmonic distortion shall not exceed 2.0% between 300 and 3000 Hz when tested at the voltages specified in 3.5.4. Under low voltage conditions, 12 to 22 volts, the distortion shall not exceed 12% between 300 and 3000 Hz.

* 3.5.8 Amplifier stability. The amplifier shall be stable and show no evidence of regeneration or oscillation when operated with input or output circuits terminated in the required impedance. No regeneration or oscillation shall occur when the power source is varied between the limits of 12 to 30 volts DC, no damage shall result from amplifier operation with input and/or output terminated in either open or short circuits.

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3.5.9 Overload protection. The amplifier shall not be damaged by overvoltage resulting from a short circuit of 390 ohms ± 5% resistance of the power source when tested In accordance with 4.7.10 (see Figure 2). The amplifier shall resume normal operation when the voltage is returned to operating limits. The amplifier shall meet the performance requirements of 3.5.2.1 upon removal of the overload condition.

* 3.5.10 Undervoltage protection. The amplifier shall not be damaged by DC bias voltages below the minimum specified in 3.5.4.2 and shall automatically resume normal operation as specified in 3.5.2.4 when the DC bias voltage returns within the limits specified in 3.5.4.

3.5.11 Voltage output. The amplifier shall be capable of providing a minimum of 0.9 volt RMS at 1000 Hz with less than 10% total harmonic distortion.

3.5.12 Internal noise. Noise level at the output shall not exceed 5 millivolt true RMS when tested in accordance with 4.7.13.

* 3.5.13 Input current. The amplifier shall require less than 20.0 mA DC when operated in accordance with Figure 2.

* 3.5.14 Electromagnetic environment (EME). The audio amplifier shall be designed, analyzed, installed, and tested to perform its function in the worldwide electromagnetic environment (EME) with out any malfunction or degradation of performance.

* 3.5.14.1 Electromagnetic compatibility (EMC). The audio amplifier shall perform its individually designed functions in a common Electromagnetic Environment (EME) with out causing or suffering unacceptable degradation due to Electromagnetic Interference (EMI) to or from other equipment/systems in the same environment in accordance with MIL-E-6051. EMC managerial guidelines are provided by MIL-HDBK-237. EMC design guidelines are provided by NAVAIR AD 1115.

* 3.5.14.2 Electromagnetic interference (EMI). The generation of EMI by the audio amplifier and the susceptibility of the audio amplifier to EMI shall be in accordance with the design limits of MIL-STD-461 and as modified by this specification. These requirements shall be met in all operating modes.

* 3.5.14.3. Electromagnetic emissions. The electromagnetic emissions requirements of MIL-STD-461 are not applicable to the audio amplifier.

* 3.5.14.4 Electromagnetic susceptibility. The fully hardened audio amplifier shall be capable of surviving exposure to the expected conducted and radiated EME in accordance with MIL-HDBK-235. As a minimum the audio amplifier electromagnetic radiated and conducted susceptibility requirements shall be in accordance with MIL-STD-461 CS114 and RS103.

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* 3.5.14.4.1 Electromagnetic susceptibility criteria. The following criteria shall be applied to determine if the audio amplifier is susceptible:

The audio output of the amplifier shall be monitored by use of an appropriate signal analyzer. The interference induced distortion in the audio signal shall not be greater than 12%.

3.6 Burn-in. The manufacturer shall subject each unit in the production lot to a burn-in test as specified in 4.7.21.

3.7 Environmental. The amplifier shall meet all Group A and Group B inspection requirements after being subjected to all environmental conditions specified below.

3.7.1 Temperature-altitude. The amplifier shall meet the applicable requirements of 3.5.2.1, 3.5.2.2 and 3.5.7 at the operating conditions specified in 3.5.4 when tested In accordance with 4.7.22.1.

3.7.2 Salt fog. After completion of the salt fog test of 4.7.22.2, the amplifier shall show no evidence of corrosion, and shall meet the requirements of 3.5.2.1 and 3.5.7 at the operating conditions specified in 3.5.4.

3.7.3 Shock test. The amplifier shall meet the requirements of 3.5.2.1 and 3.5.7 at the operating conditions specified in 3.5.4 when tested in accordance with 4.7.22.3.

3.7.4 Humidity. The amplifier shall meet the requirements of 3.5.2.1 and 3.5.7 at the operating conditions specified in 3.5.4 when tested in accordance with 4.7.22.4.

3.7.5 Thermal shock. The amplifier shall meet the requirements of 3.5.2.1 and 3.5.7 at the operating conditions specified in 3.5.4 when tested in accordance with 4.7.22.5.

3.8 Marking. The amplifier shall be marked in accordance with MIL-STD-1285. Only the following information shall be included:

Military Part Number (Type designator optional)
Contractor and Government Entities (CAGE)
Contract No.
Serial No.

3.9 Workmanship. The amplifier shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, functioning and appearance. Cleaning shall be in accordance with MIL-STD-454, Requirement 9. There shall be no evidence of loose contacts, poor or improper molding or fabrication, damaged or improperly assembled contacts, peeling, flaking or chipping of plating or finish, mechanical damage due to testing environments, nicks or burrs of metal parts of surfaces, improper or incorrect marking, improper tinning of solder cups, terminals, pins, or contacts.

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4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Qualification inspection (see 4.4).

Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraphs of this specification.

4.3.1 Ambient conditions. Unless otherwise specified, the amplifier shall be tested under the following conditions:

Temperature	Room ambient (25 ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

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4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.4) on sample units produced with equipment and procedures normally used in production. The qualification inspection shall consist of all tests specified in Table I. The qualification inspection sample shall consist of twelve units of each amplifier for which qualification is desired. Group A and B inspections shall be performed on each of the twelve sample items. At least two of the twelve units shall be considered control samples. In the event of a single failure, one of the control samples shall be submitted to the failed test sequence. If a second failure occurs, qualification shall be denied.

4.4.1 Contractors not having laboratory testing facilities satisfactory to the Government for sampling tests shall engage the services of a commercial testing laboratory acceptable to the Naval Air Warfare Center, Aircraft Division, Code 4.5.1.2.2, Indianapolis, IN.

TABLE I. Qualification inspection.

Inspection	Requirement Paragraph	Test Method
Pin stress 1/	3.4.3.1	4.7.23
Gain under standard conditions 1/	3.5.2.1	4.7.2
Gain under temperature-altitude conditions 1/	3.5.2.2	4.7.3
Gain under low voltage conditions	3.5.2.3	4.7.4
Gain stabilization time	3.5.2.4	4.7.5
Impedance 1/	3.5.3	4.7.6
Linearity	3.5.5	4.7.7
Distortion 1/	3.5.7	4.7.8
Amplifier stability	3.5.8	4.7.9
Overload protection	3.5.9	4.7.10
Undervoltage protection	3.5.10	4.7.11
Voltage output 1/	3.5.11	4.7.12
Internal noise	3.5.12	4.7.13
Input current	3.5.13	4.7.14
Electromagnetic environment (EMC)	3.5.14 2/	4.7.15 2/
Temperature-altitude 1/	3.7.1	4.7.22.1
Salt fog 1/	3.7.2	4.7.22.2
Shock 1/	3.7.3	4.7.22.3
Humidity 1/	3.7.4	4.7.22.4
Thermal shock 1/	3.7.5	4.7.22.5
Marking	3.8	4.7.24
Workmanship	3.9	4.7.25

1/ Minimum tests for QPL retention (see 4.4.4).
2/ And all sub paragraphs

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4.4.2 Prior to submitting qualification samples, the manufacturer shall have determined from actual Table I tests, and shall have submitted a test report as evidence thereof, that the products conform to the specification. Samples for qualification must be separately packaged and forwarded to the activity responsible for qualification as designated in the letter of authorization. The qualification samples shall in all respects be representative of the manufacturer's production item and shall meet the requirements of this specification. However, the amplifiers shall not have been subjected to environmental tests. Samples will be plainly identified by securely attached durable tags marked with the following information:

Identify as Qualification Sample
Military specification/specification sheet
Nomenclature (optional)
Military Part Number

Manufacturer's Part Number
Name of Manufacturer
Authorization Reference

Example

Sample for Qualification
MIL-A-23595C/3
AM-7067/A
M23595/3-1
XXXXXXX
****

Submitted (date) under authorization of (reference letter authorizing the submission)

* 4.4.2.1 Test procedures. A list of test equipment used and the procedures used for conducting all required tests shall be prepared by the contractor. The right is reserved by the Government to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved tests are available from previous contracts, such procedures will be provided and may be used. However, the right is reserved by the acquiring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.4.3 Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance of the finished product shall be upon a lot basis after satisfactory completion of all required tests. Acceptance shall be as specified in 4.5.1.

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* 4.4.4 Retention of qualification. Contractors wishing to remain on the Qualified Products List shall submit to the preparing activity at intervals no greater than three years, DD Form 1718, Certification of Qualified Products. In addition, six (6) qualification samples (see 4.4.2) shall be submitted to the QPL agent for verification of design requirements. As a minimum, the QPL agent should verify the tests noted in Table I to allow retention of qualification, and any other tests deemed necessary to insure that the basic design requirements for the amplifier are still being met. The contractor should submit production units, or units representing production as Qualification samples.

4.4.5 First article inspection. For amplifiers not covered by specification sheets or not qualified at the time set for the opening of bids First Article Inspection shall be performed. After award of the contract and prior to production, a First Article Inspection shall be performed at a laboratory acceptable to the qualifying activity (NAWCADIND). First Article Inspection shall be identical to that required for qualification.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all amplifiers of the same type, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in Table II. All amplifiers shall be inspected.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in Table III and shall be made on sample units which have been subjected to and have passed Group A inspection.

TABLE II. Group A inspection.

Inspection	Requirement Paragraph	Test Method
Gain under standard conditions	3.5.2.1	4.7.2
Impedance	3.5.3	4.7.6
Linearity	3.5.5	4.7.7
Burn-in (manufacturer performed)	3.6	4.7.21
Marking	3.8	4.7.24
Workmanship	3.9	4.7.25

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TABLE III. Group B inspection.

Inspection	Requirement Paragraph	Test Method
Gain under low voltage conditions	3.5.2.3	4.7.4
Distortion	3.5.7	4.7.8
Amplifier stability	3.5.8	4.7.9
Overload protection	3.5.9	4.7.10
Voltage output	3.5.11	4.7.12
Internal noise	3.5.12	4.7.13
Input current	3.5.13	4.7.14

4.5.1.3.1 Sampling plan for for Group B inspection. The sampling plan shall be in accordance with Table IV. If one sample fails the lot is rejected. Contracts for less than ten amplifiers shall be subject to 100% inspection.

TABLE IV Sampling Plan for Group B Inspection.

NUMBER OF AMPLIFIER ASSEMBLIES IN CONTRACT	TOTAL NUMBER OF AMPLIFIERS TO BE TESTED
10 THRU 280	3
281 THRU 500	6
501 THRU 1000	9
1001 THRU 2000	12
2001 THRU 5000	15
5001 OR MORE	18

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed the Group B inspection may be delivered on the contract or purchase order if the lot is accepted and the deliverable sample units are still within specified electrical tolerances.

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4.5.1.3.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found by lot, contract number and serial number, and the method of correcting them.

* 4.6 Deleted.

4.7 Methods of inspection. The following, methods of inspection shall be used to verify the performance requirements of 3.5 and 3.7. All electrical performance inspections shall be made with the mating connectors (U-172, U-173 or U-179 as appropriate) connected to the amplifier input and output pins. The following methods of inspection shall be performed using the test circuit of Figure 2 except that a true RMS AC voltmeter (see 6.7.7.1), which is calibrated to the 50 microvolt level, may be substituted for the lock-in amplifier. Unless otherwise specified, the voltage shall be as specified in 3.5.4 and the amplifier output shall be terminated with a non-inductive high precision (±1%) resistive load equivalent to the nominal load impedance. The input voltage to the amplifier is provided by the shielded attenuator circuit (Figure 3), which provides 10,000 to 1 attenuation (80 dB). The input voltage to the amplifier under test shall be determined by measuring the input voltage to the calibrated attenuator and dividing by 10,000. Thus, a function generator output level of 2.5 volts will provide the nominal 250 microvolts to the amplifier under test. Any variation on the test circuit of Figure 2 must be approved by the Qualifying Activity (Naval Air Warfare Center Aircraft Division, Code 4.5.1.2.2) prior to acceptance of the test results.

4.7.1 Set-up procedure. Use the test circuit described in Figure 2. Before the amplifier under test is connected, set switch S201 so that R201 Is out of the circuit (S201 shall remain in this position throughout the test sequence until the impedance test procedure of 4.7.6 is performed). Adjust the signal generator for a 2.5 volt signal into the shielded calibrated attenuator at 1000 Hz. Connect the amplifier under test and perform tests specified below.

4.7.2 Voltage gain under standard conditions. The voltage gain of the amplifier shall be determined using the test circuit of Figure 2. The output voltage shall be measured using a spectrum analyzer (see 6.7.7.3) or true RMS voltmeter (see 6.7.7.1 or 6.7.7.6), and the voltage gain calculated in dB using the following formula:

$$\text{Gain (dB)} = 20 \log (\text{Vout/Vin})$$

As a minimum, the gain shall be measured at 100, 200, 300, 400, 1000, 2000, 3000, 4000, 6000, 10,000 and 100,000 Hz to insure conformance to the requirements of 3.5.2.1.

4.7.3 Gain under temperature-altitude conditions. The voltage gain shall be determined at 1000 Hz while using the test circuit in Figure 2 and while being subjected to the environmental conditions of 4.7.22.1. The voltage gain shall conform to the limits specified in 3.5.2.2.

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4.7.4 Gain under low DC voltage conditions. The voltage gain shall be determined at 1000 Hz using the test circuit in Figure 2. The gain shall conform to the low voltage requirements specified in 3.5.2.3.

4.7.5 Gain stabilization time. The gain stabilization time shall be measured to ensure compliance with the requirements of 3.5.2.4.

4.7.6 Impedance. Impedance shall be measured at 1000 Hz and shall meet the requirements of 3.5.3 when tested with the test circuit of Figure 2. Set switch S201 so that R201 is out of the circuit. Set the signal generator to 1000 Hz, 500 microvolt RMS signal at the input of the amplifier under test. Set switch S201 so that R201 (see 6.7.7.5) is included in the test circuit. Adjust R201 so that the amplifier input voltage drops to 250 microvolts RMS. The value of R201 shall be recorded as the amplifier input impedance.

4.7.7 Linearity. The amplifier shall meet the linearity requirements of
3.5.5 when tested with the test circuit of Figure 2. The voltage gain of the amplifier shall be measured at 1000 Hz with the amplifier input voltage initially set to 50 microvolts and then slowly increased to 500 microvolts for each set bias voltage.

4.7.8 Distortion. The total harmonic distortion (THD) shall be measured at fundamental frequencies of 300, 1000 and 3000 Hz to insure compliance with 3.5.7 when tested with the test circuit of Figure 2. All harmonics within the range of 300 to 6000 Hz shall be considered. The total harmonic distortion (THD) of the amplifier under load conditions shall be determined using a distortion analyzer (see 6.7.7.4) or by using a spectrum analyzer (see 6.7.7.3) to determine the amplitudes of the harmonics by using the following equation:

THD (Total Harmonic Distortion) = [the square root of $\Sigma(Ai/Af)^2$] x 100%

Where A_i is the amplitude of the ith harmonic and A_f is the amplitude of the fundamental.

4.7.9 Amplifier stability. The amplifier THD shall be checked for conformance with the stability requirements of 3.5.8. Using the test circuit of Figure 2, the DC supply voltage shall be varied from 12 to 30 volts while measuring THD, and the load shall be removed and then restored to verify that no damage occurs. Amplifier stability need only be checked at 1000 Hz.

4.7.10 Overload protection. The amplifier shall be tested for conformance with the requirements of 3.5.9. The amplifier shall be capable of withstanding the overload condition for fifteen minutes.

4.7.11 Under voltage protection. The amplifier shall be tested for conformance with the requirements of 3.5.10.

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4.7.12 Voltage output. Using the test circuit of Figure 2, the input voltage shall be increased until a .9 volt RMS output signal is obtained. The distortion shall be measured to insure conformance with 3.5.11.

4.7.13 Internal noise. With the audio input terminals short circuited and a DC bias of 22 to 30 volts applied to the amplifier, the noise shall be measured at the amplifier output terminals with a voltmeter satisfying the requirements of 6.7.7.1. The noise voltage shall conform to requirements of 3.5.12.

4.7.14 Input current. The power supply shall be adjusted so that 27.5 volts DC appears across the output terminals of the amplifier under nominal load conditions at 1000 Hz. The input current shall be checked with a DC current meter.

* 4.7.15 Electromagnetic environment tests. The amplifier shall be tested in accordance with MIL-STD-462 and meet the requirements of 3.5.14.

* 4.7.15.1 Electromagnetic environment test plan and report. The contractor shall perform the EMI tests in accordance with an approved EMI test plan. The test plan shall describe all EMI tests required to demonstrate compliance of the equipment with the requirements of 3.5.14 and shall address all anomalies, inconsistences, and omissions. The test procedure must be approved by the qualifying activity prior to the commencement of the EMI test effort. The test report shall be forwarded to the qualifying activity. (See 6.3).

* 4.7.15.2 Electromagnetic emissions. The electromagnetic emissions tests are not applicable to the audio amplifier.

* 4.7.15.3 Electromagnetic susceptibility tests.

* 4.7.15.3.1 Conducted susceptibility. The audio amplifier power/signal cable shall be tested for conducted susceptibility in accordance with MIL-STD-462 Test Method CS114. The induced cable current shall be in accordance with curve 5 of figure CS114-1 of MIL-STD-461 to an upper frequency limit of 30 MHz.

* 4.7.15.3.2 Radiated susceptibility. The audio amplifier shall be tested for radiated susceptibility in accordance with MIL-STD-462 Test Method RS103. The RS103 field levels shall be 200 V/m with modulation format and scan rates as specified in MIL-STD-462. The qualification tests shall be performed in accordance with figures 6 and 6A. The test frequency range shall be 2 MHz to 2GHz.

4.7.16 Deleted.

4.7.17 Deleted.

4.7.18 Deleted.

4.7.19 Deleted.

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4.7.20 Deleted.

4.7.21 Burn-in. Each unit in the production lot shall be subjected to a burn-in test consisting of temperature cycling for 24 hours. The unit shall be operational during burn-in. A cycle shall consist of 1 hour at -54°C and 1 hour at 55°C. The rate of change shall be a minimum of 5°C per minute. At the completion of the 24 hour period the unit shall meet the Group A inspection requirements and shall be capable of meeting the requirements of 3.5.

4.7.22 Environmental tests.

4.7.22.1 Temperature-altitude. The amplifier shall be tested in accordance with MIL-STD-810, Method 520.0, Procedure III, except that humidity and vibration environments shall be deleted. Delete Steps 2 through 7 and substitute the following:

a. Step 2. With the amplifier non-operating, adjust the chamber test conditions to -62°C and ambient pressure. The amplifier temperature shall be stabilized and maintained for at least 2 hours.

b. Step 3. With the amplifier non-operating, adjust the chamber temperature to -54°C. The amplifier shall be turned on and the altitude adjusted to 50,000 feet. Upon reaching the specified altitude, the amplifier shall meet the requirement of 3.7.1 at the highest input voltage specified in 3.5.4.

c. Step 4. With the amplifier non-operating, adjust the chamber test conditions to -10°C and ambient pressure. After the amplifier temperature has stabilized, the chamber door shall be opened and frost permitted to form on the amplifier. The door shall remain open long enough for the frost to melt, but not long enough to allow the moisture to evaporate. The chamber door shall be closed and the amplifier turned on and shall meet the requirements of 3.7.1 at the highest input voltage specified in 3.5.4. The amplifier shall be turned on and off at least three times.

d. Step 5. With the amplifier non-operating, adjust the chamber test conditions to ambient temperature and pressure. After the amplifier temperature has stabilized, the amplifier shall meet the requirements of 3.7.1 at the highest input voltage specified in 3.5.4.

e. Step 6. With the amplifier non-operating, adjust the chamber test conditions to +85°C and ambient pressure. The chamber temperature shall be stabilized and maintained for at least 16 hours.

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f. Step 7. With the amplifier non-operating, adjust the chamber conditions to +55°C and ambient pressure. After the chamber conditions and the amplifier temperature have stabilized, turn the amplifier on at the highest input voltage specified in 3.5.4 and permit it to operate continuously for 4 hours. At the end of the operating period, and still at the specified chamber conditions, the amplifier shall be tested to determine that it meets the requirements of 3.7.1.

g. Step 8. With the amplifier non-operating, adjust the chamber temperature to +35°C and ambient pressure. After the amplifier temperature has stabilized, the amplifier shall be turned on at the highest input voltage specified in 3.5.4 and the altitude shall be adjusted to 50,000 feet. Following altitude stabilization, the amplifier shall be operated at the highest input voltage for four 30-minute time periods. Each of the four periods shall be followed by a 15 minute off period. The amplifier shall meet the requirements of 3.7.1 during the first and last periods of operation and the results recorded.

h. Step 9. With the amplifier operating, adjust the chamber conditions to ambient temperature and pressure. Following stabilization, the amplifier shall meet the requirements of 3.7.1.

4.7.22.2 Salt fog. The amplifier shall be tested in accordance with MIL-STD-810, Method 509.2, Procedure I. The following details apply:

a. Step 2. Following completion of the 48 hour exposure, the amplifier shall be removed from the test chamber and operated within 2 hours. The amplifier shall meet the requirements of 3.7.2.

b. Step 3. The drying period shall be 72 hours.

c. Step 4. The amplifier shall meet the requirements of 3.7.2.

4.7.22.3 Shock. The amplifier shall be tested in accordance with MIL-STD-810, Method 516.3, Procedure I, except as follows:

a. The shock spectrum shall be a standard half-sine pulse configuration in accordance with Figure 5 of this specification.

b. The amplifier shall be subjected to three shocks applied in both directions along each of three mutually perpendicular axes. Functional testing shall be performed prior to and after application of all shocks.

c. Functional testing shall be in accordance with 3.7.3.

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4.7.22.4 Humidity. The amplifier shall be tested in accordance with MIL-STD-810, Method 507.2, Procedure III, except as follows:

a. Step 2. The chamber temperature shall be 71°C.

b. Step 4. Maintain 85 percent or greater relative humidity and reduce internal chamber temperature in 16 hours to 28° ± 10°C (82° ± 18°F).

c. Step 5. Delete.

d. Step 6. Do not repeat Step 5.

e. Step 7. The amplifier shall be operated at the end of the tenth cycle only. Within one hour following completion of the tenth cycle, the amplifier shall be operated and shall meet the requirements of 3.7.4.

4.7.22.5 Thermal shock. The amplifier shall be tested in accordance with MIL-STD-810, Method 503.2, Procedure I, except as noted below. The temperature extremes shall be -54°C and +71°C, and the stabilization time shall be 15 minutes at each temperature extreme.

a. Step 5. Repeat Step 2.

b. Step 7. Following completion of Step 6, the amplifier shall meet the requirements of 3.7.5.

4.7.23 Pin stress test. A snug-fitting pin block simulating the mating connector shall be attached to the amplifier output pins and a stress to breakage test, typical of but not limited to that illustrated in Figure 4, shall be performed. The applied force rate shall be no less than 1 pound per second up to but not exceeding a force of 45 pounds. At the completion of the test, the amplifier shall meet the requirements of 3.4.3.1, 3.5.2.1 and 3.5.7.

4.7.24 Marking. The amplifier shall be inspected to insure conformance with 3.8.

4.7.25 Workmanship. The amplifier shall be inspected to insure conformance with 3.9 and 3.3.12.

4.7.26 Fungus. The amplifier shall be inspected in accordance with MIL-STD-810, Method 508.3. Amplifiers shall meet the requirements of Table 508.3-I, Grades 0 or 1, and shall be evaluated using data of I-4.1.

4.7.27 Flammability. Amplifiers shall be tested in accordance with ASTM D635-81 to determine if the requirements of 3.3.4 have been met. Amplifiers shall be considered nonflammable if the following conditions are met:

a. Item is self extinguishing.

b. Item does not produce flames, smoke, toxic fumes.

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5. PACKAGING

5.1 Packaging. All units shall be preserved, packaged and packed for the level of shipment specified in the contract or purchase order in accordance with MIL-E-17555.

5.2 Marking. Unless otherwise specified in the contract or purchase order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The amplifiers covered by this specification are intended for use in any application where their performance characteristics are required. The amplifiers are suitable for installation on aircraft systems within the limitations of applicable performance requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Applicable detail specification number, title and date.

c. Applicable detail specification part number.

d. Number of amplifiers required.

e. Level of packaging and packing required.

f. Responsibility for inspection, if other than specified (Section 4).

g. Special preparation for delivery requirements, if applicable (Section 5).

* 6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.7.15.1	DI-EMCS-80201A	ELECTROMAGNETIC INTERFERENCE TEST PROCEDURES	1/
4.7.15.1	DI-EMCS-80200A	ELECTROMAGNETIC INTERFERENCE TEST REPORT	2/

1/ The EMI Test Procedure shall describe all the tests required to demonstrate compliance with the requirements of 3.5.14. The EMI Test Procedure shall detail the equipment used to monitor the audio amplifier during the testing as well as all associated instrumention.The EMI Test Procedure shall be approved by the qualifying activity prior to the start of EMI testing. The contractor shall perform the EMI tests in accordance with the approved EMI Test Procedure.

2/ The contractor shall supply the results of the EMI test effort in an approved EMI Test Report. The EMI Test Report shall be approved by the qualifying activity prior to the qualification acceptance of the audio amplifier. The EMI Test Report shall describe all of EMI tests performed to demonstrate compliance with the requirements of 3.5.14 and shall address all anomalies, inconsistences, and omissions, (if any), of the EMI Test Procedure.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.3.1 Conformity to qualified sample. It is understood that amplifiers supplied under contract shall be identical in every respect to the qualification sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

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6.4 Application for qualification. The activity responsible for the QPL for MIL-A-23595(AS) is the Naval Air Systems Command. The Naval Air Warfare Center, Indianapolis, IN, has been designated by the Naval Air Systems Command as agent for the establishment of the QPL. Request for information pertaining to and applications for qualification should be addressed to:

Commanding Officer
Naval Air Warfare Center
Code 4.5.1.2.2
6000 East 21st Street
Indianapolis, IN 46219-2189

6.5 Inspection system requirements. The manufacturer shall maintain a system of inspection in conformance with MIL-I-45208. Test equipment shall be calibrated at intervals of no greater than one year.

6.6 Subject term (keyword) listing.

Amplifier
Audio amplifier
In-line amplifier

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.7 Rationale. The following historical documentation provides rationale for various requirements contained in this specification.

6.7.1 Pin stress. The pin stress force values of 45 pounds (3.4.3.1) reflect the force values prevalent in current Government procured amplifiers that have not been the subject of Quality Deficiency Reports (QDRS) for pin breakage.

6.7.2 Signal input voltage. The nominal signal input voltage of 250 microvolts rms (3.5.4.1), and the linearity voltage range of 50 to 500 microvolts rms (3.5.5) simulates the signal output voltage of the M-87/AIC microphones.

6.7.3 DC bias voltages. The nominal bias voltage operating range of 22 to 30 volts DC (3.5.4.2) and the low voltage conditions of 12 to 22 volts DC (3.5.2.3) are based upon the aircraft system power voltages called out in MIL-STD-704.

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6.7.4 Radiated susceptibility, electric field. The maximum electric field strength of 200 volts per meter (3.5.16) is taken from MIL-STD-461 requirement for aircraft. If the aircraft must see that field strength then so should the aircraft systems and components. In addition, pilots have reported audio buzzing signals when encountering a radar sweep.

6.7.5 Radiated susceptibility, magnetic field. The maximum magnetically induced voltage of 15 millivolts rms (3.5.16), between 200 Hz to 100 KHz, is based upon the minimum threshold voltage necessary to switch on the AIC-14 Voice Activated Relay (VOX). This threshold voltage switches on the AIC-14 VOX even if outside the audio frequency range.

6.7.6 Environmental tests. The listed MIL-STD-810 environmental tests (see 3.7.1, 3.7.2, 3.7.3, 3.7.4, 3.7.5) have been tailored to reflect the previous requirements of MIL-T-5422. MIL-T-5422 was made inactive for new design. However, this amplifier specification has been tailored to more accurately describe the amplifier that the Government has been procuring for many years. Semantically speaking, the amplifier is a "new design" and requires invoking MIL-STD-810. However, the intent of this specification is not to require the redesign of the existing amplifier. Therefore, MIL-STD-810 had to be tailored to reflect the previous requirements of MIL-T-5422.

6.7.7 Test equipment. The test equipment described herein was selected to minimize testing problems encountered in measuring low level AC true RMS voltages and impedances. The low level AC microvoltages are several orders of magnitude lower in value than the thermally induced voltages. The low level impedances are equivalent to short circuits for some instruments and thereby create test induced measurement errors. Although automated test equipment is permitted and encouraged, be aware of test induced problems created by various analog to digital conversion techniques and related problems. These test induced problems include the generation of nonexistent failures as well as the concealment of bonafide failures.

6.7.7.1 Multimeter.

a. Digital readout (4-1/4 digits)

b. Direct current (DC) and true RMS alternating current (AC)

c. Ranges: 10 microvolts to 15 volts RMS AC 1/
1 to 30 volts DC

d. Accuracy: ±.5%

e. Input impedance: 1 megohm minimum

f. Inputs isolated from ground up to 30 volts

g. Frequency response: 100 Hz to 100 KHz

1/ NOTE: A calibrated lock-in amplifier with fixed incremental gain control may be used as a true RMS AC voltmeter provided all other requirements are met.

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6.7.7.2 Signal or function generator.

a. Sine wave output

b. Frequency range: 100 Hz to 100 KHz minimum

c. Frequency accuracy and flatness at full output: ±2 percent

d. Harmonic distortion: Less than -65 dB (relative to fundamental) at full output for each frequency

e. DC offset: 1 millivolt maximum

f. Voltage range: 1 millivolt to 15 volts 2/

g. Output isolated from ground up to 30 volts

2/ NOTE: A suitable amplifier may be used providing the function generator plus amplifier combination meet the above requirements.

6.7.7.3 Spectrum analyzer (optional).

a. Frequency range: 100 Hz to 100 KHz

b. Center frequency accuracy: ±.003%

c. Bandwidth: 1 Hz minimum to 500 Hz maximum

d. Measurement ranges: 1 microvolt to 15 volts linear
-120 dBV to +30 dBV log

e. Dynamic range: .70 dB

f. Voltage accuracy: ±10 volts AC RMS

g. Input impedance: I megohm minimum

h. Inputs isolated from ground up to 30 volts

6.7.7.4 Distortion analyzer.

a. Frequency range: 100 Hz to 100 KHz

b. Frequency accuracy: ±2%

c. Measurement range: 30 microvolts to 15 volts

d. Dynamic range: 70 dB

e. Voltage accuracy: ±.5%

f. Input impedance: .1 megohm minimum

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g. Fundamental rejection: 70 dB

h. Inputs isolated from ground up to 30 volts

6.7.7.5 Precision decade resistance box.

a. Frequency range: DC to 100 KHz

b. Resistance range: 2 to 15 ohms

c. Resistance increments: .1 ohms

d. Accuracy: .5%

e. Resistance zero setting: 12 milliohms or less

f. Inductance: Less than .03 microhenries

g. Capacitance: Less than 8 picofarads

h. Low temperature and power coefficients

6.7.7.6 Lock-in amplifier (optional).

a. Frequency range: 100 Hz to 100 KHz

b. Sensitivity: 1 microvolt to .5 volts

c. Output time constants: 1 millisecond to 10 seconds

d. Harmonic rejection: 55 dB

e. Normal output stability: 50 ppm

f. Gain accuracy: ±2%

g. Digital readout (3-1/2 digits)

h. Phasing: Automatic

i. Input Impedance: 1 megohm or greater

j. Gain: Incremental or lockable after calibration

6.7.7.7 Oscilloscope.

a. Channel: Single

b. Bandwidth: 5 megahertz

c. Time base: 20 microsec/div to 1 millisec/div

d. Time base accuracy: ±3%

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e. Coupling: AC and DC

f. Source: Internal

g. Sweep mode: Auto trigger

h. Voltage base: 5 millivolts/div to 10 volts/div

i. Voltage base accuracy: ±3%

j. Input impedance: 1 megohm minimum

6.7.7.8 AC ratio transformer standard (see 6.8).

a. Ratio: 1.111111 maximum, 0.0111111 minimum

b. Frequency range: 50 Hz to 10 kHz

c. Terminal linearity: 0.0001% (1 ppm)

d. Resolution: 0.00001% steps

e. Maximum effective impedance: R series 3.5 ohm, L series 100 uH

6.7.8 Figure 2. The output circuitry of Figure 2 simulates the AIC-14 intercommunication system.

6.7.9 Figure 3. The amplifier input voltage range of 50 to 500 microvolts RMS creates NBS traceability problems for calibrated AC true RMS voltmeters. Many multimeters and true RMS voltmeters have unwarranted claims as to an AC true microvoltage measuring capability. On the lowest range scale, the instruments do not maintain the measurement nor calibration accuracy that is true at higher range scales. The instrument specification sheet itself is often unclear as to the actual accuracy when measuring a microvolt signal.The attenuation circuit of Figure 3 can resolve this problem without the use of an expensive AC true RMS microvoltmeter provided the following conditions are met:

a. The attenuation value is calibrated with NBS traceability at measurable voltage levels.

b. The impedances do match between the function generator and the amplifier under test.

c. The attenuation value is 80 dB and not 60 dB.

NOTE: EXPERIMENTAL EVIDENCE SHOWS THAT A 60 DB ATTENUATOR DOES NOT PROVIDE A CONSTANT VOLTAGE INDEPENDENT OF FREQUENCY OR ANTICIPATED LOAD IMPEDANCES.

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An AC ratio transformer standard (see 6.7.7.1), whose ratio is calibrated at a higher voltage level, can serve as an adequate transfer standard for verifying an instrument with AC true RMS microvoltage measuring capability and for calibrating the attenuator described in Figure 3.

6.7.10 Figure 5. The shock test acceleration value of 15 g's simulates anticipated values that occur when the amplifier is dropped.

Preparing activity:
Navy - AS
(Project No. 5831-N095)

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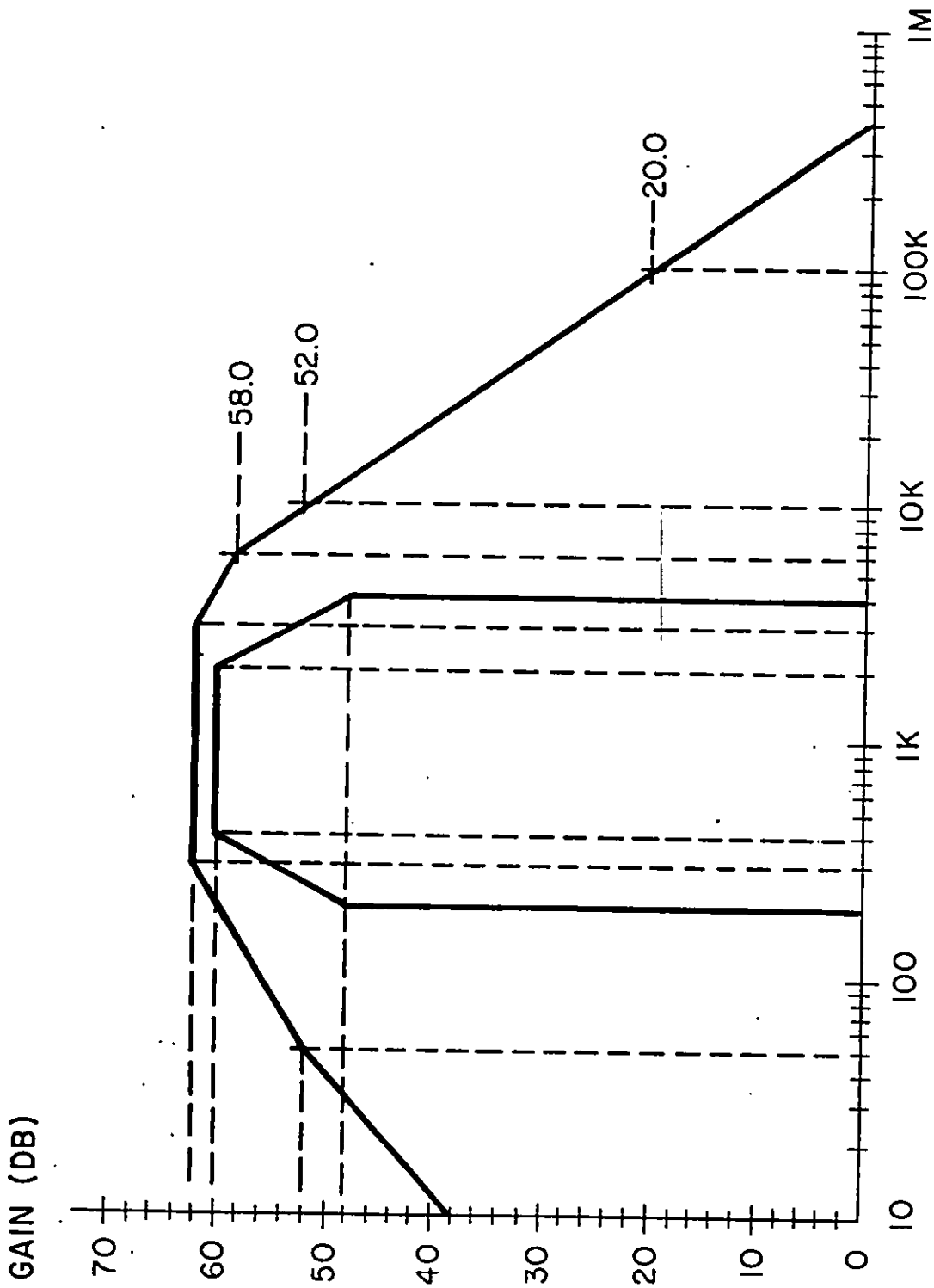
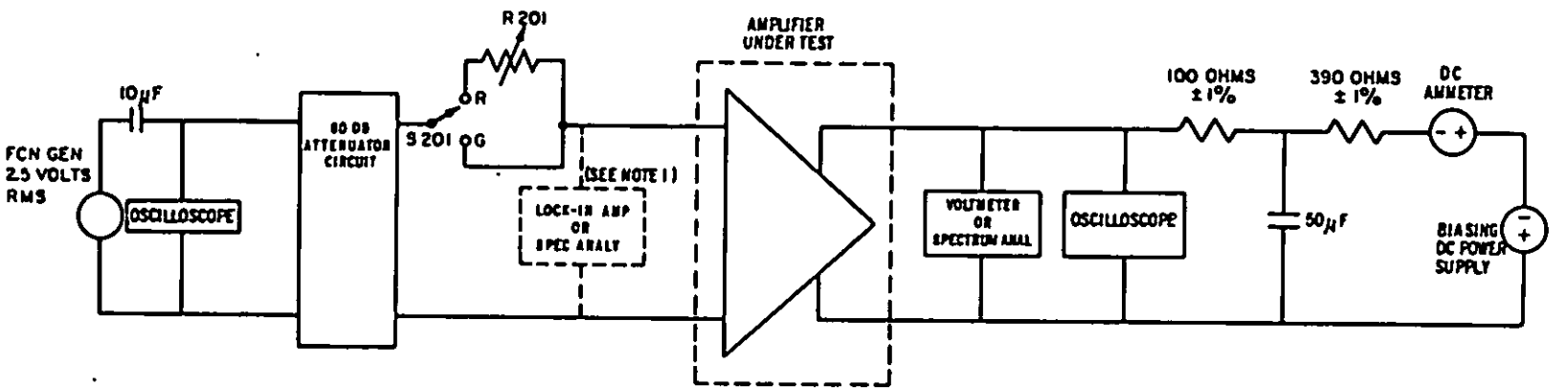


FIGURE 1. Amplifier frequency response curve.

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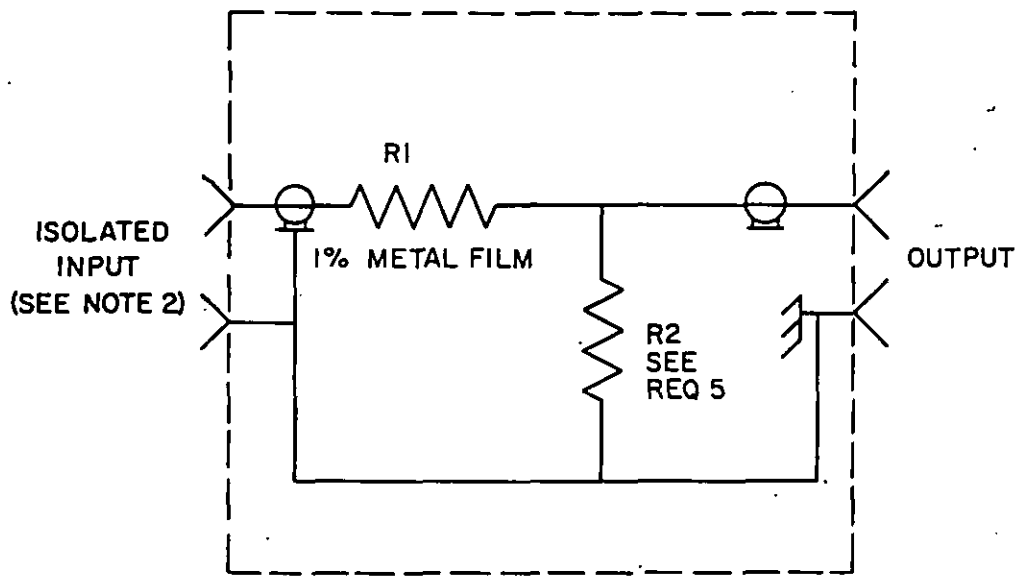
MIL-A-23595D(AS)

NOTES:

1. Neither the lock-in amplifier nor the spectrum analyzer is required if the attenuator is calibrated.
2. Oscilloscope is required to ascertain if any waveshape distortion, oscillation, etc., is present.
3. R201 = Decade resistance box IAW 4.3.1.5.
4. Switch S201 shall be placed in the "R" position when performing the amplifier input impedance test, and placed in the "G" position when performing all other tests. S201 is not necessarily a physical switch, but depicts a break for inserting R201.
5. The output circuitry simulates the AIC-14 intercommunication system.

FIGURE 2. Amplifier test circuit.

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NOTES:

1. Case shall be Pomona Electronics, Inc., Model 3320 or equivalent.

2. The existing input connector shall be replaced with isolated BNC type 9 Amphenol P/N 31-10 or equivalent.

3. Resistor R1 nominally equals [(A-1)/A] * Zs, where A is a 10,000 to 1 attenuation, and Zs is the function generator output impedance.

4. Resistor R2 nominally equals $(Zs * Z_L)/((A * Z_L) - Zs)$, where A is a 10,000 to 1 attenuation, Zs is the function generator output impedance, and Z_L is the nominal load impedance.

5. Resistor R2 shall be a nominal one inch length of manganin resistance wire, of suitable AWG size, and shall be calibrated to provide 10,000 to 1 attenuation. For 50 ohm function generator output, use .08926 ohm/foot Driver-Harris AWG #15, .05707" dia, or equivalent. For 600 ohm function generator output, use .5678 ohm/foot Driver-Harris AWG #23, .02257" dia, or equivalent (see 6.8 for calibration background).

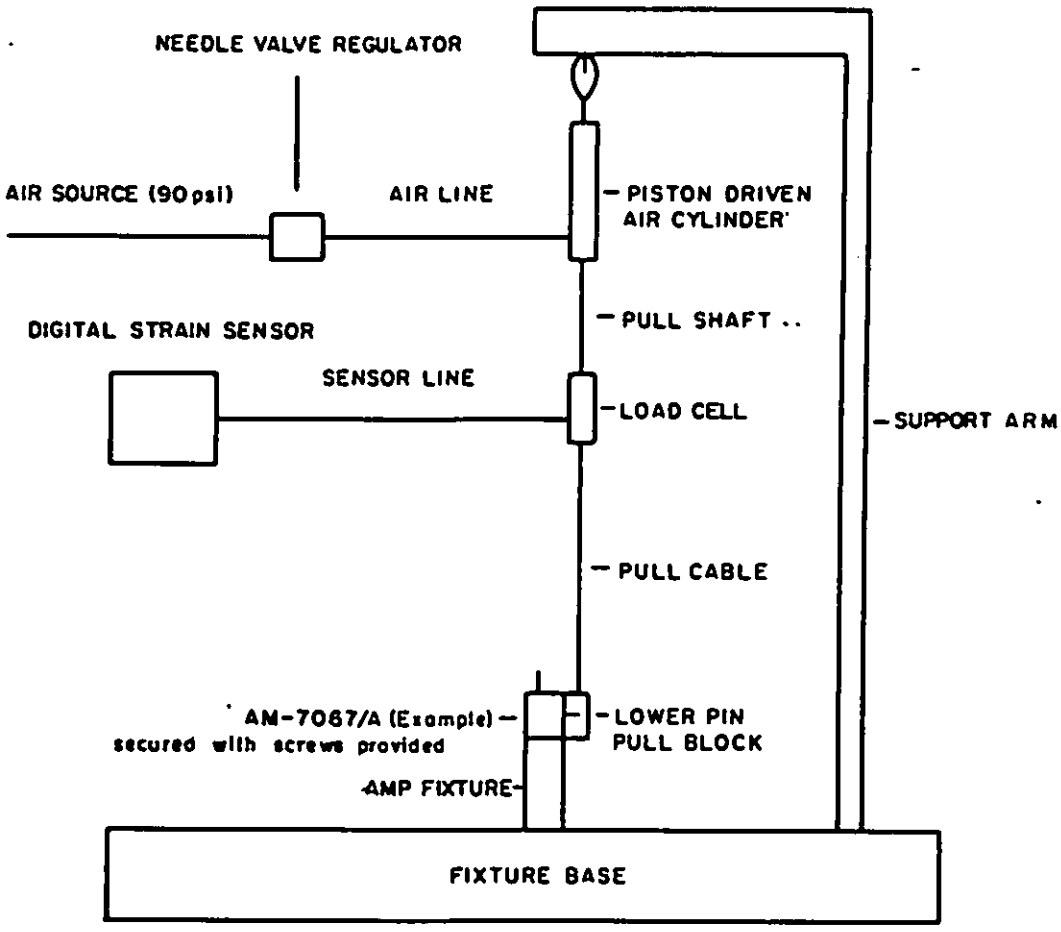
6. A commonly available resistance may be used for R1, provided (R1 + R2)/R2 is 10,000 (80 dB).

7. The attenuator is described in literature as a fixed asymmetrical L pad attenuator.

8. NOTE: Experimental evidence shows that a 60 dB attenuator does not provide a constant voltage that is independent of frequency or anticipated load impedances.

FIGURE 3. Attenuator schematic.

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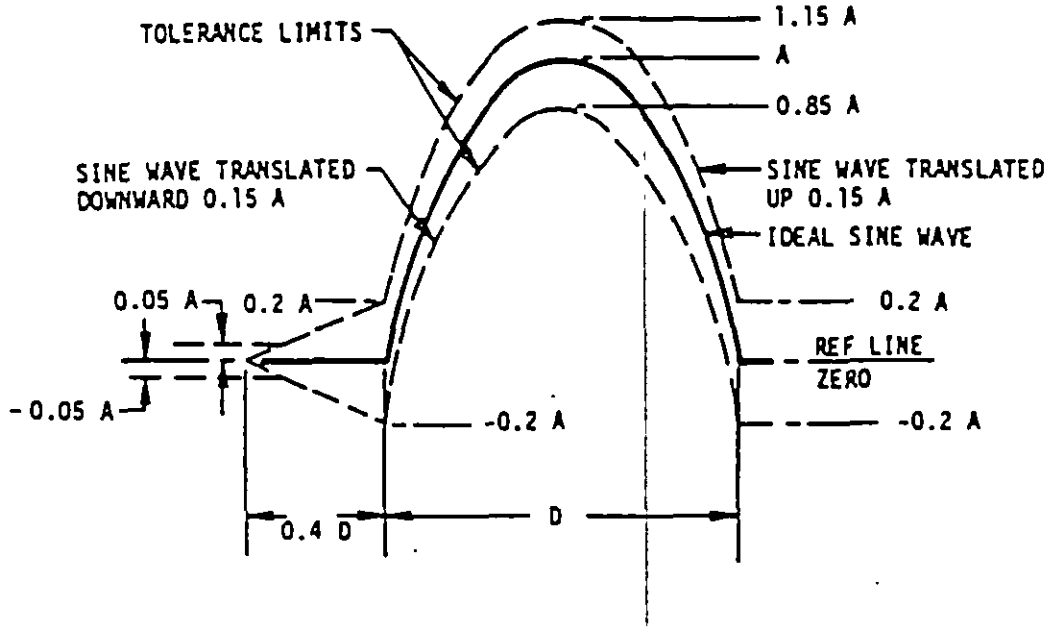


NOTE:

This figure illustrates a typical test configuration only. Any equivalent test method is acceptable to achieve the end results as described in MIL-A-23595(AS) step 3.4.3.1.

FIGURE 4. Typical amplifier pin stress to breakage test configuration.

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ACCELERATION (A), g	DURATION (D), ms
15	11 ± 1

The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is acceptable.

FIGURE 5. Half sine shock pulse configuration and its tolerance limits.

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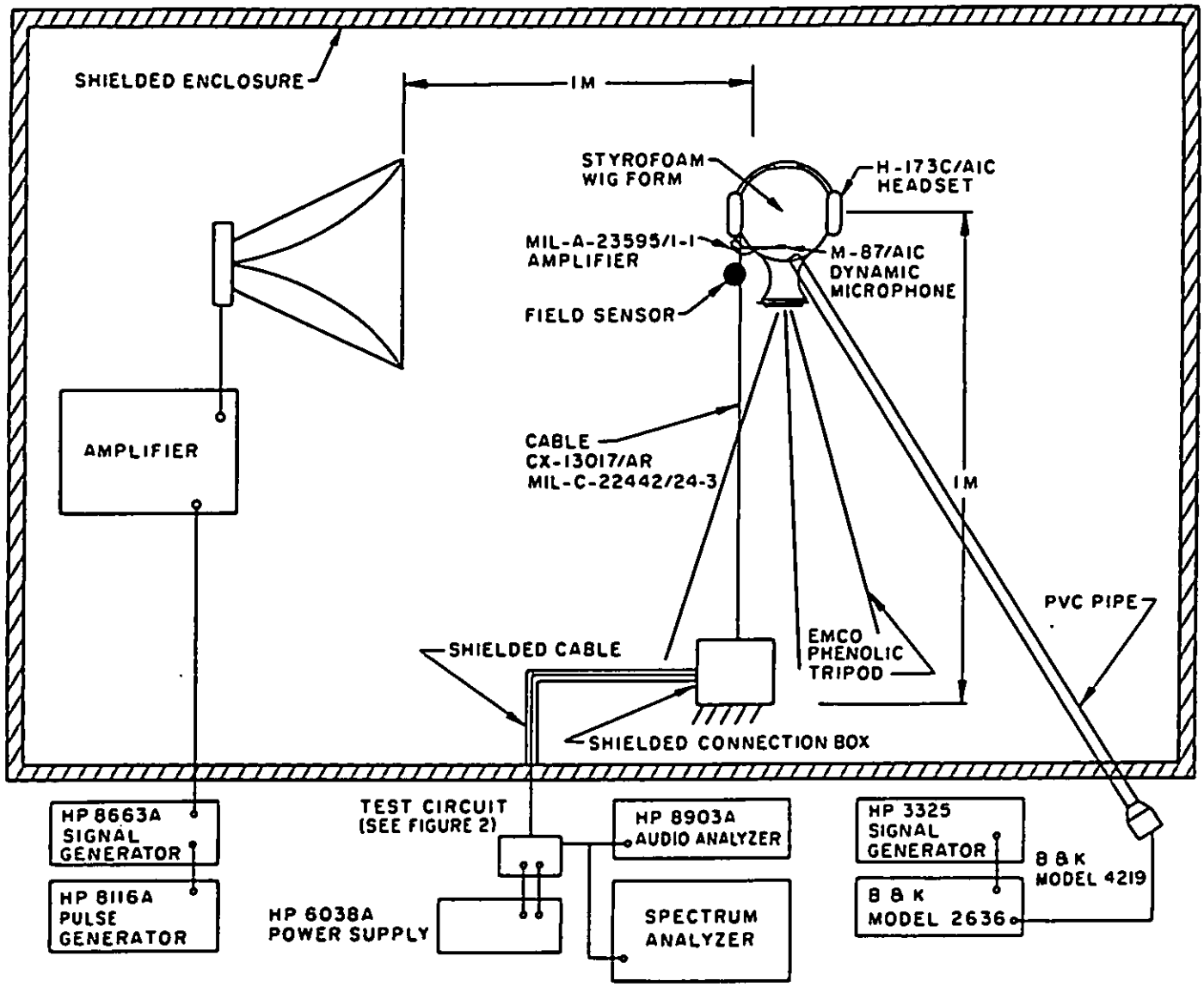


FIGURE 6 RADIATED SUSCEPTIBILITY TEST SET UP

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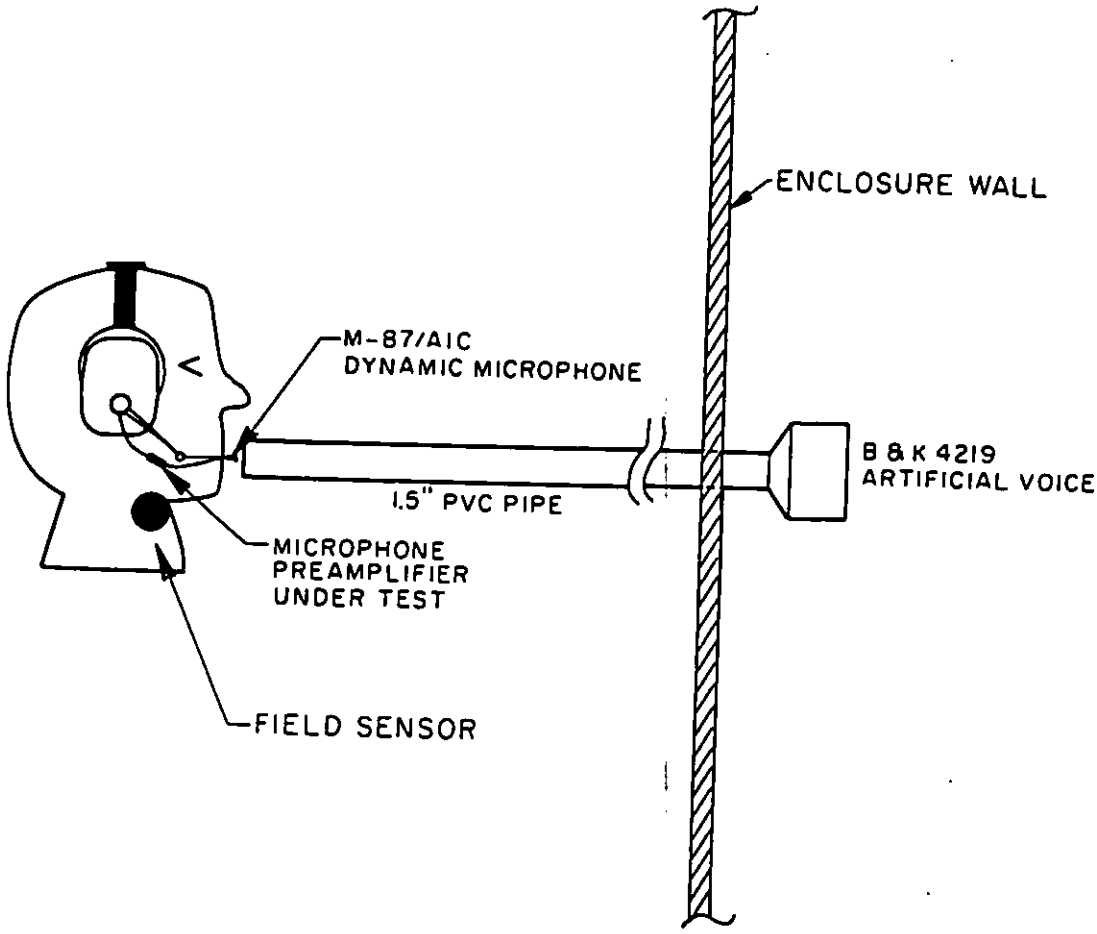


FIGURE 6A RADIATED SUSCEPTIBILITY TEST SET UP (ARTIFICAL VOICE INPUT DETAIL)

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